آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 13

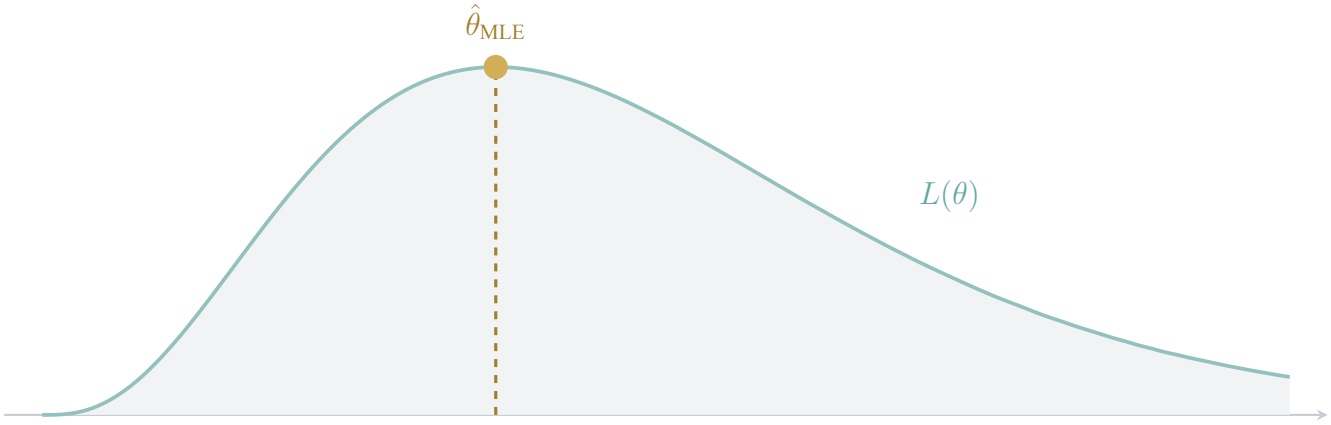
Parameter Estimation

برآورد پارامتر

Instructor: Amin Azari

Compiled by: Aidin Shekari

استاد: امین آذری github.com/AminAzari



نیمسال پاییز ۱۴۰۶-۱۴۰۵
Fall Semester 2026-2027

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Learning Objectives

After completing this module, students will be able to:

1. Distinguish between estimator (rule) and estimate (number)
2. Evaluate estimator properties: bias, consistency, efficiency, MSE
3. Apply the Method of Moments
4. Apply Maximum Likelihood Estimation (MLE)
5. Compare estimators using MSE and bias-variance tradeoff

13.1 Introduction

We observe data $X_1, \ldots, X_n$ from a distribution with unknown parameter θ. **Parameter estimation** gives us a method to infer θ from data.

Engineering examples:

- Estimate noise variance σ^2 from recorded samples
- Estimate failure rate λ from component lifetimes
- Estimate channel gain from received signal strength
- Estimate mean signal level from measurements

13.2 Estimator vs. Estimate

Definitions

Term	Type	Definition	Example
Estimator $\hat{\theta}$	Random variable (function of data)	The RULE applied to data	$\hat{\theta} = \bar{X} = \frac{1}{n}\sum X_i$
Estimate	Number	The VALUE from a specific sample	$\hat{\theta} = 3.47$ (from one dataset)

Key Insight

The **estimator** is a random variable — it has a sampling distribution.
The **estimate** is one realization of that random variable.

Analogy: "Take the average" is the estimator (rule). "The average was 3.47" is the estimate (result).

13.3 Properties of Estimators

Bias

$$B(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- **Unbiased:** $B(\hat{\theta}) = 0$, i.e., $E[\hat{\theta}] = \theta$ (on average, correct)
- **Biased:** $E[\hat{\theta}] \neq \theta$ (systematic error)

Examples:

- $\bar{X}$ is unbiased for μ: $E[\bar{X}] = \mu$ ✔
- S^2 (with $n-1$) is unbiased for σ^2: $E[S^2] = \sigma^2$ ✔
- S (sample std dev) is biased for σ: $E[S] \neq \sigma$ (slight downward bias)

Consistency

$\hat{\theta}_n$ is **consistent** if $\hat{\theta}_n \to \theta$ in probability as $n \to \infty$.

Practical meaning: with enough data, the estimator gets arbitrarily close to the truth.

Sufficient condition: If bias $\to 0$ and variance $\to 0$ as $n \to \infty$, then consistent.

Efficiency

Among all unbiased estimators, the **efficient** one has the smallest variance.

The **Cramér-Rao Lower Bound** (CRLB) gives the minimum possible variance:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

where $I(\theta)$ = Fisher information $= -E\left[\frac{\partial^2 \ln f(X;\theta)}{\partial \theta^2}\right]$.

An estimator achieving the CRLB is called **efficient** or **minimum variance unbiased (MVU)**.

Mean Squared Error (MSE)

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [B(\hat{\theta})]^2$$

$\text{MSE} = \text{Variance} + \text{Bias}^2$

This is the **bias-variance tradeoff**: sometimes a slightly biased estimator with much lower variance has smaller MSE than an unbiased one.

13.4 Method of Moments (MoM)

Principle

Set population moments equal to sample moments, then solve for parameters.

- $k-$ th population moment: $\mu' k = E[X^k]$
- $k-$ th sample moment: $m' k = \frac{1}{n}\sum X_i^k$

Procedure

1. Express parameters in terms of population moments
2. Replace population moments with sample moments
3. Solve for parameter estimates

Example: Exponential Distribution

Data from Exp(λ). Population moment: $E[X] = \frac{1}{\lambda}$.
Set equal: $\frac{1}{n}\sum X_i = \frac{1}{\hat{\lambda}} \rightarrow \hat{\lambda}_{\text{MoM}} = \frac{1}{\bar{X}}$

Example: Gaussian Distribution

Data from $N(\mu, \sigma^2)$:

- 1st moment: $\hat{\mu} = \bar{X}$
- 2nd central moment: $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ (note: biased, uses n not $n-1$)

Advantages/Disadvantages

✔ Simple, always applicable, closed-form solutions often available
✘ Not always efficient, can produce biased estimates, may give invalid parameter values

13.5 Maximum Likelihood Estimation (MLE)

Principle

Choose the parameter value that makes the observed data **most probable**.

Likelihood Function

Given data $x_1, \ldots, x_n$ from $f(x;\theta)$:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

Log-Likelihood (usually easier)

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i;\theta)$$

MLE Procedure

1. Write the log-likelihood $\ell(\theta)$
2. Differentiate: $\frac{d\ell}{d\theta} = 0$
3. Solve for $\hat{\theta}_{\text{MLE}}$
4. Verify it's a maximum $\left(\frac{d^2\ell}{d\theta^2} < 0\right)$

Properties of MLE

1. **Consistent** (converges to true value)
2. **Asymptotically unbiased** (bias $\to 0$ as $n \to \infty$)
3. **Asymptotically efficient** (achieves CRLB for large n)
4. **Invariant:** If $\hat{\theta}$ is MLE of θ, then $g(\hat{\theta})$ is MLE of $g(\theta)$

Example: MLE for Gaussian Mean

Data $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known.

$$\ell(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$$

$$\frac{d\ell}{d\mu} = \frac{1}{\sigma^2}\sum(x_i - \mu) = 0 \to \hat{\mu}_{\text{MLE}} = \bar{X}$$

Example: MLE for Exponential Rate

Data $X_1, \ldots, X_n \sim \text{Exp}(\lambda)$.

$$\ell(\lambda) = n \cdot \ln(\lambda) - \lambda \cdot \sum x_i$$

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum x_i = 0 \to \hat{\lambda}_{\text{MLE}} = \frac{n}{\sum x_i} = \frac{1}{\bar{X}}$$

Example: MLE for Gaussian Variance

Data $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, μ known.

$$\ell(\sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$$

$$\frac{d\ell}{d(\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(x_i - \mu)^2 = 0$$

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum(x_i - \mu)^2$$

Note: With μ unknown, $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum(x_i - \bar{x})^2$ — biased! (Uses n, not $n-1$)

13.6 Engineering Examples

Estimating Noise Variance

Problem: Measure n samples of noise. Estimate noise power σ^2.

Data: $x_1, \ldots, x_n$ (noise voltage samples, assumed zero-mean)

MLE: $\hat{\sigma}^2 = \frac{1}{n}\sum x_i^2$ (biased by factor $\frac{n-1}{n}$)
Unbiased: $S^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ or if $\mu = 0$ known: $\frac{1}{n}\sum x_i^2$ is actually unbiased for $E[X^2] = \sigma^2$

Estimating Failure Rate

Problem: 20 components tested until failure. Lifetimes: $t_1, \ldots, t_{20}$.

Model: $T \sim \text{Exp}(\lambda)$, parameter λ (failure rate).

MLE: $\hat{\lambda} = \frac{20}{\sum t_i} = \frac{1}{\bar{T}}$

If $\bar{T} = 500$ hours: $\hat{\lambda} = \frac{1}{500} = 0.002$ failures/hour.

Estimating Channel Parameter

Problem: Rayleigh fading channel. Observed power samples: $p_1, \ldots, p_n$.

Model: Power $P \sim \text{Exp}\left(\frac{1}{\Omega}\right)$, where $\Omega = E[P]$ = average power.

MLE: $\hat{\Omega} = \bar{P} = \frac{1}{n}\sum p_i$

13.7 Comparing Estimators

Bias-Variance Tradeoff

Two estimators for σ^2:

- $\hat{\theta}_1 = S^2$ (unbiased, variance $= \frac{2\sigma^4}{n-1}$)
- $\hat{\theta}_2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ (biased by $-\frac{\sigma^2}{n}$, variance $= \frac{2\sigma^4(n-1)}{n^2}$)

$$\text{MSE}(\hat{\theta}_1) = \frac{2\sigma^4}{n-1}$$

$$\text{MSE}(\hat{\theta}_2) = \frac{2\sigma^4(n-1)}{n^2} + \frac{\sigma^4}{n^2} = \frac{\sigma^4(2n-1)}{n^2}$$

For any $n \geq 2$: $\text{MSE}(\hat{\theta}_2) < \text{MSE}(\hat{\theta}_1)$! The biased MLE actually has smaller total error.

13.8 Practice Problems

Problem 1

Data from $N(\mu,\,9)$: $\bar{x} = 4.2$, $n = 25$.

(a) Give the MLE of μ. (b) What is $\text{Var}(\hat{\mu})$? (c) Is $\bar{X}$ efficient for μ?

Solution

(a) $\hat{\mu}_{\text{MLE}} = \bar{X} = 4.2$

(b) $\text{Var}(\bar{X}) = \dfrac{\sigma^2}{n} = \dfrac{9}{25} = 0.36$

(c) CRLB for $N(\mu,\,\sigma^2)$: $\frac{1}{n \cdot I(\mu)} = \frac{\sigma^2}{n} = 0.36$. $\bar{X}$ achieves CRLB $\rightarrow$ yes, efficient.

Problem 2

Component lifetimes (hours): 120, 350, 200, 480, 90, 560, 310, 175, 420, 280.

(a) Estimate λ (failure rate) by MLE assuming $\text{Exp}(\lambda)$. (b) Estimate MTTF.

Solution

(a) $\bar{X} = \dfrac{120 + 350 + 200 + 480 + 90 + 560 + 310 + 175 + 420 + 280}{10} = 298.5.$

$\hat{\lambda} = \frac{1}{298.5} = 0.00335$ failures/hour.

(b) MTTF $= \frac{1}{\hat{\lambda}} = \bar{X} = 298.5$ hours.

Problem 3

For Poisson(λ) data: $x_1, \ldots, x_n$. Derive the MLE of λ.

Solution $\ell(\lambda) = \sum[x_i \ln(\lambda) - \lambda - \ln(x_i!)] = (\sum x_i)\ln(\lambda) - n\lambda - \sum \ln(x_i!)$

$\dfrac{d\ell}{d\lambda} = \dfrac{\sum x_i}{\lambda} - n = 0 \rightarrow \hat{\lambda}_{\text{MLE}} = \dfrac{1}{n}\sum x_i = \bar{X}.$

The MLE for Poisson rate is the sample mean.

Next Module: Module 14 — Confidence Intervals